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journal homepage: www.elsevier.com/locate/ipmEnhancing neural topic modeling for social media text via semantic bag of word clusters and log-domain Sinkhorn transport[☆]Yi Sun, Junhao Zhao, Haoran Xu, Ronghua Zhang, Changzheng Liu^{ID}*, Limengzi Yuan^{ID}*

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ABSTRACT

Topic modeling has been widely applied to analyze text data from social media platforms. Under this scenario, traditional Neural Topic Models (NTMs) encounter three primary challenges: (1) initial text representation; (2) the long-tail nature of topic distributions in social network texts; (3) approximation of Optimal Transport. Motivated by these challenges, we propose an end-to-end solution spanning from text representation to topic modeling.

First, we propose SBoWC, a novel text representation method that performs dimensionality reduction while absorbing semantic information through base terms, achieved by combining word embeddings with clustering statistics. Subsequently, we propose GSWTM, a Wasserstein-based autoencoder topic model that fits the long-tail topic distribution in social network texts via Gamma priors and innovatively employs log-domain Sinkhorn to approximate Optimal Transport.

Ablation studies demonstrate the transferability and effectiveness of SBoWC in text representation. GSWTM demonstrates significantly better performance than baselines in TU, C_V , and the comprehensive metrics TQ across four real social network datasets of varying sizes. The log-domain Sinkhorn approximation exhibits excellent stability, allowing the regularization parameter ϵ to be reduced to 0.1–0.01, thereby approaching the original Optimal Transport.

1. Introduction

In the digital era, real-time, information-centric social media platforms have witnessed rapid and extensive growth. These platforms not only draw large numbers of users but also generate massive amounts of textual data, including news articles, user-generated content, comments, and advertisements. Such data has become a critical resource for a wide range of natural language processing (NLP) tasks. Topic modeling (Blei et al., 2010), as one of the crucial tasks in natural language processing, aims to extract low-dimensional latent topic structures (Zhuang et al., 2017) from massive text data and uncover the semantic information and automatically identify the key topics in the data and their distributions, providing powerful support for tasks such as sentiment analysis (Dai et al., 2024; Sufi & Khalil, 2024; Yi et al., 2025), event detection (Costa et al., 2024; Soomro et al., 2024; Wang, Zhang, & Wu, 2024; Yang et al., 2025), public opinion monitoring (Miao et al., 2023; Rkia et al., 2024; Tingjiang et al., 2023), and automatic recommendation (Liang et al., 2024; Yu et al., 2025).

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Document Representation Challenges: High-quality representations determine the performance upper bounds of machine learning algorithms (Bengio et al., 2013). A good document representation should, on the one hand, be able to faithfully reflect the content of the document, and on the other hand, possess the ability to distinguish different documents. However, common methods such as BoW and TF-IDF are widely recognized for ignoring semantic information and the linguistic structure between words in natural language processing (NLP) tasks. This sparsity and semantic poverty in traditional text representations fundamentally limits topic model performance. In this unsupervised manner, the document representation provided to LDA or similar topic models does not yield highly predictive results (Xing et al., 2012).

Model Architecture Limitations: VAE/WAE-based approaches represent the mainstream paradigm in current neural topic models (NTMs). However, most methods employ traditional prior distributions such as Dirichlet, Gaussian, or Gaussian mixture distributions (Chen et al., 2024; Das et al., 2015; Tran et al., 2020; Zhou et al., 2022). Nevertheless, these symmetric priors fail to capture the inherent asymmetry in real-world topic distributions, particularly the long-tail phenomenon prevalent in social media texts. Moreover, relying on these restrictive distributional distributions not only limits the model's ability to handle imbalanced topic intensity texts, but also imposes mutual constraints with approximate optimal transport methods, such as Maximum Mean Discrepancy (MMD) (Gretton et al., 2006). Beyond these limitations, existing approximate OT solutions themselves (Choi et al., 2025; Knop et al., 2020; Liu et al., 2022; Tolstikhin et al., 2019) also face challenges in balancing accuracy with efficiency and maintaining stable optimization, particularly when dealing with sparse or highly imbalanced distributions common in social media contexts.

In today's social media environment, discussions tend to concentrate on a few rare and surprising topical events, resulting in a long-tail distribution of topics (Agarwal et al., 2021; Peng et al., 2022). A few core topics occupy the main information space, while a large number of secondary topics are scattered with low frequency. This phenomenon has been discovered and leveraged by some studies (Wang et al., 2015; Wu et al., 2021).

To address long-tail distribution characteristics in social media text topics and overcome the aforementioned challenges, we propose an end-to-end framework that combines text representation learning with robust topic modeling:

1. We propose a theoretically principled document representation method, SBoWC (Semantic Bag of Word Clusters), which enhances the density and semantic capability of document representation through word embeddings and word clusters. Ablation experiments validate the transferability of the SBoWC text representation method across different downstream tasks and its effectiveness in improving the performance of mainstream topic models.

2. We adopt Gamma prior to capture the long-tail nature of topics in social texts. We propose the Gamma Sinkhorn Wasserstein Topic Model (GSWTM), a novel method based on the Wasserstein distance. GSWTM relies on the excellent geometric awareness of log-domain Sinkhorn algorithm to perform approximate optimal transport computations. This strategy effectively ensures numerical stability throughout the entire optimization process.

3. Experimental results on multiple real-world social media text datasets of varying sizes conclusively demonstrate that GSWTM achieves significantly superior topic coherence and diversity than all baselines in the vast majority of cases. The log-domain Sinkhorn's ϵ parameter remains stable even at very small values. Systematic ablation studies validate the effectiveness of the proposed components.

Our work unifies semantic clustering (SBoWC) and GSWTM into a coherent mechanism. Theoretically, SBoWC transforms sparse word co-occurrence signals into dense semantic cluster representations. By absorbing low-frequency word semantics while maintaining high-frequency word importance, the document representation aligns with the long-tail structure modeled by the Gamma prior. This alignment ensures the revealed long-tail semantics remain compatible with the asymmetric geometry captured by the Sinkhorn optimal transport process.

This paper is organized as follows: Section 2 reviews important related work relevant to this study; Section 3 provides a detailed introduction to the construction of the SBoWC document representation method; Section 4 describes the design of the GSWTM topic model; Section 5 conducts extensive experiments and comprehensive ablation studies based on real social media data; finally, Section 6 summarizes the entire paper and suggests future research directions. Table 1 provides a simple example that previews our contributions and explains their advantages.

2. Related work

2.1. Text representation in natural language processing (NLP) tasks

NLP tasks typically entail transforming raw textual data into numerical representations, followed by the application of computational models to accomplish specific linguistic objectives.

BoW (Salton, 1989) is a simple and intuitive method that maps documents to fixed-length vectors based on binary term presence. However, it fails to capture semantic importance and contextual information. Fig. 1 provides a simple yet illustrative example of these limitations.

To illustrate the deficiencies of the BoW model more intuitively, consider two short sentences. If the vocabulary contains the following basis terms: "car", "vehicle", "parked", "stored", "garage", "shed", then the BoW representations of these two sentences will be:

Sentence 1: "The car is parked in the garage". $\rightarrow$ Vector [1, 0, 1, 0, 1, 0]

Sentence 2: "The vehicle is stored in the shed". $\rightarrow$ Vector [0, 1, 0, 1, 0, 1]

Table 1
Comparison of topic modeling approaches for social media text analysis.

Challenge	Existing methods	Limitations	GSWTM (Ours)	Advantages
Text Representation	BOW, TF-IDF, and pre-trained embeddings (<i>FastTopic</i> , <i>W-LDA</i> , <i>BERTopic</i> , <i>CWTM</i>)	• Hard semantic matching • Loss of interpretability • Lack of probabilistic grounding	SBoWC : Semantic Bag-of-Word Clusters with fuzzy membership: $\mu_{v_i}(u) = \max(\cos(\mathbf{V}_{v_i}, \mathbf{V}_u), 0)$	• Semantic-Aware Soft Matching • Low-Frequency Term Preservation • Interpretability-Efficiency Balance
Prior Distribution	Symmetric distributions: Dirichlet (<i>W-LDA</i> , <i>TNTM</i>), Gaussian (<i>VaGTM</i> , <i>MGD</i> , <i>NSEM-GMHTM</i>)	• Poor modeling of long-tail distributions • Overfitting to head topics • Inflexible prior assumptions	Gamma (α, β) with α, β from extensive pre-experiments	• Asymmetric modeling of long-tail sparsity and intensity imbalance • Geometry-aware compatibility with Sinkhorn transport
OT Approximation	MMD(<i>CWTM</i> , <i>W-LDA</i>), KL(<i>BigARTM</i>), Sinkhorn(<i>vONTSS</i> , <i>NSTM</i> , <i>ASTM</i>)	• Numerical stability • Limiting Distributional Assumptions • Accuracy-Efficiency Trade-off	Log-domain Sinkhorn : $\sum_{i,j} \exp(\log u_i + \log v_j - C_{ij}/\epsilon) C_{ij}$	• Preservation of geometric structure • Numerical overflow/underflow elimination • Superior approximation with small ϵ

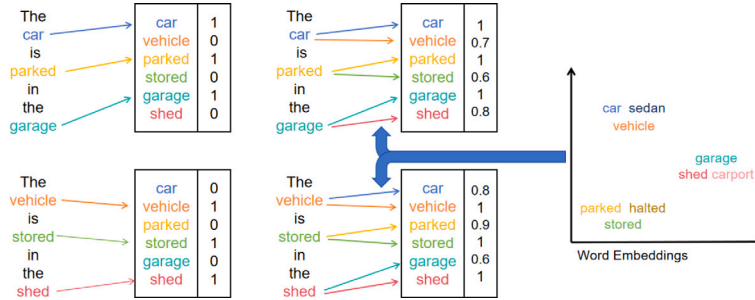


Fig. 1. Comparison of BoW representations for semantically similar sentences.

Although the two sentences are semantically similar, BoW fails to capture this, resulting in significantly different vector representations.

Although many later flat representation methods (Blei et al., 2003; Liu et al., 2015; Nguyen et al., 2015; Wu et al., 2017) effectively reduced the high dimensionality of BoW, they still failed to capture semantic associations (between “car” and “vehicle”).

Recent neural and deep learning methods have advanced text representation, but typically require prior knowledge such as supervised signals or pre-trained word embeddings. Existing approaches include using semantic similarity in probabilistic language models (Yao et al., 2024), augmenting bag-of-words with co-occurrence features (Castro et al., 2022), combining TF-IDF and GloVe (Singh et al., 2022), and integrating unsupervised with supervised learning (Talukdar & Biswas, 2024). While deep models capture word order, they face challenges like high computational cost, reliance on large labeled datasets, complex hyperparameter tuning, and limited long-range dependency modeling. These limitations can restrict their use in scenarios with constrained resources, scarce annotations, or long documents. To address this, we aim to propose a simpler, unsupervised, and efficient alternative.

Fuzzy mapping has recently emerged as a novel and effective interpretable method for capturing semantics. However, existing approaches (Akhtar et al., 2019; Muffo et al., 2023; Sijin & Champa, 2020) are not fully suitable for downstream tasks such as topic modeling, or suffer from other limitations. For instance, FBoW (Zhao & Mao, 2018) excessively compensates for sparsity, introducing significant noise, and is only applicable to classification tasks; SFBoW (Muffo et al., 2021) is constrained to generating fixed-dimensional static sentence embeddings; FBoT (Jia & Li, 2018) relies heavily on external resources and empirical parameters.

Distinctively, compared to techniques that rely on “word-pair” similarity, SBoWC constructs a fundamentally different global semantic framework. Based on the principle of “semantic absorption”, SBoWC preserves the dominance of high-frequency words while simultaneously supplementing the often-overlooked semantic value of low-frequency words. This approach uniquely addresses critical limitations such as fragmented semantics, loss of low-frequency words, and noisy redundancy while simultaneously dynamically optimizing the weight of high-frequency base terms. Consequently, the document representation aligns more closely with the true semantic structure, making it particularly suitable for downstream tasks like topic modeling, which require clear semantic signals and complete topic information.

2.2. Text clustering and topic modeling

Text clustering and topic modeling have a long history. Topic modeling can be seen as a special type of soft clustering method, where the “cluster centers” are semantically meaningful topics. Each topic is defined by a word distribution, and each document is represented by a topic distribution.

Integrating Document Clustering and Topic Modeling: MGCTM (Xie & Xing, 2013) pioneered a unified framework that jointly performs document clustering and topic modeling, thereby discovering both cluster-specific local topics and globally shared topics within a single generative model. Sun (2014) systematically categorized topic model-based document clustering methods into two groups: methods based on basic topic models and clustering-oriented topic models. They demonstrated that simple clustering methods based on basic topic models can achieve performance comparable to state-of-the-art algorithms.

In recent years, Costa and Ortale (2019) and Costa and Ortale (2020) proposed unified frameworks based on Bayesian generative models, treating document clustering and topic modeling as mutually reinforcing joint tasks. Furthermore, Costa and Ortale (2020) integrated word embedding techniques. Seifollahi et al. (2021) proposed a two-stage topic modeling algorithm that first discovers topic word distributions by softly clustering embedded n-grams, and then determines document-topic distributions through sampling. Amiri and Karimianghadim (2024) constructed documents as a network structure and combined community detection algorithms from social network analysis with topic modeling, proposing an innovative document clustering model.

Traditional topic modeling methods: Probability-based conventional methods such as LDA (Blei et al., 2003), PLSA (Hofmann, 1999), and HDP (Teh et al., 2006) once flourished. However, they typically rely on strong prior assumptions, and their inference processes often depend on approximate algorithms (such as Gibbs sampling), which are limited when processing large-scale corpora or capturing complex semantic relationships. Matrix factorization methods such as LSA (Deerwester et al., 1990) and NMF (Lee & Seung, 1999), while capable of extracting latent topics, have strong linear assumptions and lack sufficient modeling of textual semantics.

In recent years, NTMs (neural topic models) have emerged, overcoming the limitations of traditional methods by introducing the powerful nonlinear representation capabilities of neural networks (Wu, Nguyen, & Luu, 2024).

Embedding-based NTMs (George & Sumathy, 2023; Wu et al., 2023; Wu, Nguyen, Zhang, et al., 2024; Xu, Jiang, et al., 2023) represent topics as low-dimensional vectors, leveraging semantic knowledge from pre-trained word embeddings to enhance both computational efficiency and model flexibility. **Contrastive Learning-based NTMs** (Han et al., 2023; Luo et al., 2024; Nguyen et al., 2024; Zhou et al., 2023) optimize topic representations by learning similarity relationships between positive and negative sample pairs, improving semantic consistency and discriminability. **Adversarial Training-based NTMs** (Wang et al., 2022; Zhang et al., 2022) employ generator–discriminator architectures to enhance topic robustness and the authenticity of generated documents. With the rise of large language models, **Pre-trained LM-based NTMs** have also emerged: (Doi et al., 2024) alleviate sparse word co-occurrence in short texts via prompting strategies, TopicGPT (Pham et al., 2024) generates human-readable topic labels and descriptions, and DeTiME (Xu, Hu, et al., 2023) integrates encoder–decoder LMs with diffusion models for topic-guided generation.

NTMs with Variational Autoencoders remain among the widely adopted approaches in topic modeling. NVDM (Miao et al., 2016) first introduced the VAE framework to this field, enabling flexible posterior approximation via neural networks, while ProdLDA (Srivastava & Sutton, 2017a) further refined this framework with improved modeling.

Among state-of-the-art works, Topic-VQ-VAE (Yoo & Choi, 2024) incorporates vector quantization into VAEs to achieve discrete topic representations (treating latent codebooks as conceptual bags-of-words), yet its codebook learning tends to prioritize high-frequency terms and may struggle with capturing sparse semantic information in short texts. vONTSS (Xu, Jiang, et al., 2023) replaces traditional Gaussian distributions with von Mises-Fisher distributions and combines optimal transport for semi-supervised topic modeling, but it often fails to align the distribution of low-frequency terms due to limited annotated samples for such terms. FET-LM (Tu et al., 2024) employs normalizing flows based on Householder transformations to enhance sequential posterior modeling, though its performance relies heavily on sufficient text sequences—making it less effective for sparse, short social media texts. DepNTM (Huang et al., 2024) integrates structural causal models into VAEs to learn topic dependencies in directed acyclic graphs, yet it cannot reliably model dependencies involving long-tail terms.

Critically, the aforementioned existing methods either overlook sparse and discrete semantics due to the “continuous space assumption” or lose long-tail information due to “sample size dependence”, creating a significant gap in effectively addressing the characteristics of social media short texts, such as fragmented semantics and uncommon long-tail terms.

2.3. Optimal transport in topic modeling

Current mainstream topic models often involve or rely on optimal transport theory when optimizing the similarity between topic distributions, making the OT problem an unavoidable and critical challenge. In recent years, models based on Variational Autoencoders (VAE) (Kingma & Welling, 2013) and Wasserstein Autoencoders (WAE) (Tolstikhin et al., 2019) have gained significant attention. These models (Shan et al., 2025; Tang et al., 2024; Wang, Wang, et al., 2024; Yoo & Choi, 2024) leverage the expressive power of deep neural networks to capture complex topic distributions more effectively. However, their approximate OT strategies often rely on surrogate divergence measures such as KL divergence (Kullback & Leibler, 1951) or MMD (Dziugaite et al., 2015), which can lead to posterior collapse in the case of KL and impose strict architectural constraints when using MMD-based priors.

Furthermore, the flat geometry of MMD does not faithfully capture ground-truth distances, and may occasionally cause gradient vanishing near boundary regions of the support, thereby hindering stable learning (Genevay et al., 2018). Chakraborty and Bartlett (2024) demonstrated that WAEs can effectively learn data distributions when the network architecture is well-designed, with

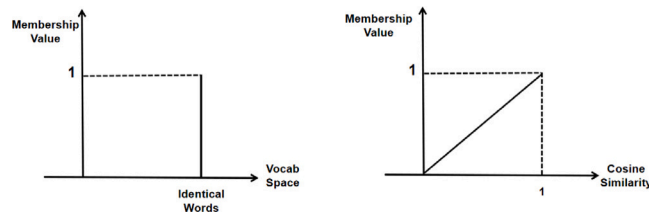


Fig. 2. Comparison of two mappings.

performance closely tied to the intrinsic dimensionality of the data. They further argued that ensuring topological compatibility between the prior and the data distribution is critical.

The Sinkhorn algorithm (Cuturi, 2013) stands out due to its excellent geometric perception capabilities. Some researchers have begun incorporating Sinkhorn distance into distribution alignment in topic models; however, certain limitations still remain. The performance of NSTM (Zhao et al., 2021) using Sinkhorn entropy regularization is limited by the quality of the input pretrained word embeddings (Niu et al., 2024). The ASTM (Liu et al., 2022) based on Sinkhorn divergence may suffer from instability in high-dimensional latent spaces and slow convergence. The LANTM (Chen et al., 2025), which integrates label guidance and optimal transport mechanisms, relies heavily on the availability of high-quality labels. It also has a relatively high model complexity and potentially large training cost. Moreover, all three models face a common issue: the entropic regularization parameter ϵ cannot be chosen arbitrarily small to approximate the true OT transport plan, due to numerical instability and computational burden.

To address these fundamental limitations, this study adopts the log-domain Sinkhorn (Schmitzer, 2019), which significantly overcomes the sensitivity to the choice of the entropic regularization parameter ϵ . It flexibly captures the geometric structure between probability distributions and the matching relationships between samples. Moreover, by using a smaller ϵ , it can better approximate the exact OT solution.

3. SBoWC

In this section, we propose the SBoWC (Semantic Bag-of-Word Clusters), an extension of the traditional bag-of-words approach. Since its membership function is constructed based on word embeddings, we first introduce the concept of word embeddings.

3.1. Word embedding

Word embedding (Bojanowski et al., 2017; Mikolov et al., 2013; Pennington et al., 2014) is a technique that maps discrete words into a continuous vector space, aiming to capture the semantic and syntactic relationships between words. Among various word embedding methods, Word2Vec (Mikolov et al., 2013) is a widely used unsupervised neural network-based model that efficiently learns vector representations of words. Thanks to its streamlined architecture, Word2Vec can be trained on large-scale unlabeled corpora and is capable of effectively encoding meaningful semantic relationships.

In this study, we adopt Word2Vec as the underlying model for generating word embeddings and use cosine similarity to measure the similarity between word vectors. Word2Vec has demonstrated strong performance and stability across a wide range of natural language processing tasks (Adewumi et al., 2022; Alkathiri et al., 2021; Hendrawan et al., 2022).

Typically, cosine similarity is used to measure the semantic similarity between word embeddings. Given two words, their embeddings are denoted as $\mathbf{A}$ and $\mathbf{B}$, respectively. A higher cosine similarity indicates a stronger semantic correlation between the words, while a lower (or even negative) value may suggest semantic opposition, such as antonymy.

The formula for calculating Cosine Similarity is as follows:

$$\text{Cosine Similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \quad (1)$$

where: $\mathbf{A}$ and $\mathbf{B}$ are vector representations of two words (or texts) (word vectors such as TF - IDF vectors are often used). $\mathbf{A} \cdot \mathbf{B}$ represents the dot product, and its calculation formula is:

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^n A_i B_i \quad (2)$$

$\|\mathbf{A}\|$ and $\|\mathbf{B}\|$ represent the Euclidean norms (L2 norms) of the vectors, and their calculation formulas are respectively:

$$\|\mathbf{A}\| = \sqrt{\sum_{i=1}^n A_i^2}, \quad \|\mathbf{B}\| = \sqrt{\sum_{i=1}^n B_i^2} \quad (3)$$

To illustrate this point, Table 2 presents the five words with the highest similarity to the example words “book” and “student”, along with their cosine similarity scores.

Cosine similarity measures the relative orientation between two word vectors by computing the cosine of the angle between them, making it invariant to differences in magnitude or frequency. This allows it to effectively capture semantic similarity: vectors

Table 2
Similar words and cosine similarity scores.

Inquiry	Similar words	Cosine similarity
Research	study	0.352
	discussion	0.346
	analysis	0.332
	science	0.310
	exploration	0.298
Education	learning	0.360
	teaching	0.345
	school	0.328
	pedagogy	0.318
	curriculum	0.317

of semantically similar words (e.g., “car” and “vehicle”) typically form small angles and yield high similarity, while unrelated words (e.g., “car” and “banana”) form larger angles and result in lower similarity scores. In addition, cosine similarity is computationally efficient, involving only dot product operations and norm normalization, which makes it well-suited for large-scale word similarity computations.

3.2. Fuzzy membership function based on word embedding

In the BoW, words in a document – after preprocessing – are exactly matched to a predefined set of basis terms, and each document is represented as a sparse binary vector via hard mapping. Specifically, if a word w matches a basis term v_i , the output of the mapping function is 1; otherwise, it is 0. This exact matching process can be interpreted as a crisp membership function defined as follows:

$$\mu_{v_i}(w) = \begin{cases} 1, & \text{if } w = v_i \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

As shown in Fig. 2, BoW represents text based solely on word occurrence. This leads to sparse document vectors that fail to capture meaning, causing semantically similar sentences with different word choices to appear unrelated. In topic modeling, such limitations reduce topic coherence and introduce redundancy. To address this, we adopt a semantic matching strategy based on word similarity, implemented via a fuzzy membership function illustrated in Fig. 2.

$$\mu_{v_i}(w) = \begin{cases} \cos(V_{v_i}, V_w), & \text{if } \cos(V_{v_i}, V_w) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

Let v_i denote a basis term and w a word to be matched, with their corresponding word embeddings V_{v_i} and V_w obtained from Word2Vec. The cosine similarity between V_{v_i} and V_w , ranging from $[-1, 1]$, is used to measure semantic similarity. Only positively correlated words (i.e., similarity > 0) are considered to enhance semantic alignment. Unlike hard membership, which only counts exact matches, the fuzzy membership function incorporates semantically similar words, significantly improving the representational capacity of the BoW model. However, the high dimensionality of the resulting vectors remains a challenge. In the following, we describe how to integrate this semantic similarity function into the BoW framework to obtain semantically enriched document representations.

3.3. Bag-of-words calculation based on word clusters

In large-scale corpora, document vectors are often high-dimensional yet sparse, as each document contains only a small subset of a vast vocabulary. To reduce sparsity and computational cost, a common strategy is to retain only the top 20%–30% high-frequency words after preprocessing as basis terms. While this captures the core vocabulary and improves efficiency, it risks omitting semantically important low-frequency words, leading to incomplete topic representations. For example, in political texts, crucial terms like “impeachment” may appear infrequently but carry significant meaning; filtering them out can result in the loss of key semantic information.

To avoid omitting semantically important low-frequency words while still achieving dimensionality reduction, we propose a weight optimization strategy based on the semantic relevance of word clusters. The theoretical foundation of this strategy lies in the assumption that the topic word set derived from modeling should, to the greatest extent possible, capture the full or core semantics of the majority of textual content within the event corpus.

After determining the set of basis terms v , we select a semantic word cluster c_i for each basis term v_i , where v_i serves as the representative of its cluster. We then calculate the semantic similarity between all words in c_i and each word in the vocabulary. All negative similarity values are set to zero. For each vocabulary word, the maximum positive semantic similarity with the words

in c_i is added to the cluster weight of the corresponding basis term v_i . This positive semantic screening enables the basis terms to absorb the semantic contributions of low-frequency words in the vocabulary that are not selected as basis terms but share similar meanings. As a result, the weights of those basis terms that effectively represent a topic are further enhanced, while the weights of weakly representative ones are slightly reduced.

$$\mu_{v_i}(w) = \text{MAX}\left(\left[\max(\cos(W[w_{i1}], W[w]), 0), \dots, \max(\cos(W[w_{iit}], W[w]), 0)\right]\right) \quad (6)$$

Let the i th word cluster be $c_i = \{w_{i1}, w_{i2}, \dots, w_{iit}\}$, where t_i is the number of words in c_i . To compute the similarity between a target word w and the cluster c_i , we calculate the cosine similarity between w and each $w_{ij} \in c_i$, and take the maximum value. This yields a similarity score in $[-1, 1]$. This operation guarantees a similarity of 1 for self-mapping, reinforces the weights of representative basis terms, and absorbs the semantic contributions of non-basis low-frequency words.

For a document d , the SBoWC model represents it as $\mathbf{x}_d = [x_{d1}, x_{d2}, \dots, x_{di}]$, where each x_{di} is the sum of the membership degrees of all words semantically aligned with the i th basis term:

$$x_{di} = \sum_{w_{\omega'} \in \mathbf{W}} \mu_{v_i}(w_{\omega'}) \cdot A(w_{\omega'}) \quad (7)$$

Here, $\mathbf{W}$ is the vocabulary, v_i is the i th basis term (cluster centroid), $w_{\omega'}$ represents the ω' -th word in the vocabulary (where $\omega' \in \{1, 2, \dots, \omega\}$ and ω is the vocabulary size), and A is a weighting function. TF-IDF can be used, where $A(w_{\omega'}) = \text{TF-IDF}(w_{\omega'})$. For clarity and simplicity, we adopt the raw counting scheme in the subsequent algorithm: $A(w_{\omega'}) = \text{count}(w_{\omega'}, d) = \beta_{d\omega'}$, the number of times $w_{\omega'}$ appears in document d .

Algorithm 1 Semantic Bag of Words Frameworks

Input: a text corpus D with k documents; the vocabulary $\mathbf{W}$ and its corresponding word embeddings matrix $\mathbf{\Omega} \in \mathbb{R}^{j \times \omega}$, where ω is the vocabulary size and j is the dimensionality of word embeddings. The number of selected high-frequency words v : i , where $i \leq \omega$

Output: learned document vectors for the corpus: $\mathbf{X} \in \mathbb{R}^{k \times i}$

1: Based on $\mathbf{W}$, obtain matrix $\mathbf{B} \in \mathbb{R}^{k \times \omega}$ where each row vector $\beta_d = [\beta_{d1}, \beta_{d2}, \dots, \beta_{d\omega}] \in \mathbb{R}^{\omega}$ represents the d -th document ($d \in \{1, \dots, k\}$), with $\beta_{d\omega'}$ counting occurrences of word $\mathbf{W}_{\omega'}$.

if SBoWC is performed **then**

2: Compute TF-IDF and PMI scores for all words, and calculate their harmonic mean as the overall importance score:

$$\text{Score}(w_{\omega'}) = \frac{2 \cdot \text{TF-IDF}(w_{\omega'}) \cdot \text{PMI}(w_{\omega'})}{\text{TF-IDF}(w_{\omega'}) + \text{PMI}(w_{\omega'})}$$

Select the top- i words with the highest scores, and extract their embeddings $\mathbf{\Omega}_I \in \mathbb{R}^{i \times j}$ from $\mathbf{\Omega}$.

3: Cluster words in $\mathbf{\Omega}_I$ using K-means with i clusters.

4: Obtain the embeddings of words in each cluster, and compute the cosine similarity (based on Eq (6)) between these cluster words and words in documents; Construct the transformation matrix $\mathbf{M} \in \mathbb{R}^{\omega \times i}$;

$$\mathbf{M} = \begin{bmatrix} \mu_{v_1}(w_1) & \mu_{v_2}(w_1) & \dots & \mu_{v_i}(w_1) \\ \mu_{v_1}(w_2) & \mu_{v_2}(w_2) & \dots & \mu_{v_i}(w_2) \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{v_1}(w_{\omega'}) & \mu_{v_2}(w_{\omega'}) & \dots & \mu_{v_i}(w_{\omega'}) \end{bmatrix} \quad (8)$$

end if

5: Let $\mathbf{B}_{\text{bin}}$ be the binary version of $\mathbf{B}$, where each entry is set to 1 if the corresponding word appears in the document (i.e., $\beta_{d\omega'} > 0$), and 0 otherwise. Then compute the semantic document representation matrix $\mathbf{X}$ as:

$$\mathbf{X} = (\mathbf{B}_{\text{bin}} \circ \mathbf{B}) \mathbf{M}$$

where $\circ$ denotes the Hadamard (element-wise) product.

return $\mathbf{X}$

In Step 2, we do not rely on simple frequency counts for basis term selection. Instead, we fuse TF-IDF and PMI scores using their harmonic mean to ensure that selected words are both statistically significant and semantically informative.

In Step 4, matrix $\mathbf{M} \in \mathbb{R}^{\omega \times i}$ serves as a semantic projection operator that maps each of the ω vocabulary words to i semantic clusters. Each element $\mu_{v_k}(w_{\omega'})$ represents the cosine similarity between word $w_{\omega'}$ and cluster center v_k , effectively encoding how strongly each word belongs to each semantic cluster. The subsequent matrix multiplication $\mathbf{X} = \mathbf{B} \mathbf{M}$ transforms document-word frequencies into document-cluster affinities, aggregating semantically related words and producing dense representations that capture semantic rather than lexical patterns.

In Step 5, we retain only the weights of basis terms that actually appear in each document. This filtering operation eliminates irrelevant information introduced by unseen basis terms, reducing semantic noise and enhancing the alignment between the document representation and its actual content. Although this increases sparsity, it helps suppress background interference and sharpens topic signals, thereby improving the quality of downstream topic modeling or clustering tasks.

Table 3

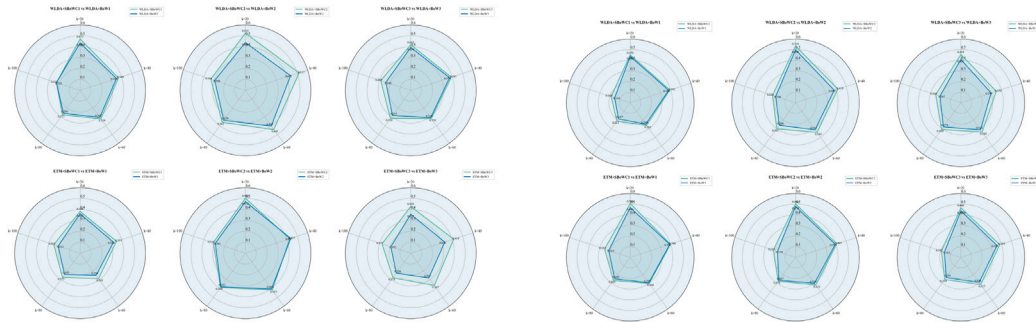
The topic model experiment on AGnews.

AGnews	$k = 20$			$k = 40$			$k = 60$			$k = 80$			$k = 100$		
metrics	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
W-LDA+SBoWC1	0.836	0.564	0.472	0.768	0.480	0.369	0.736	0.440	0.324	0.663	0.418	0.277	0.593	0.384	0.228
W-LDA+BoW1	0.819	0.531	0.435	0.762	0.459	0.350	0.734	0.418	0.307	0.652	0.399	0.260	0.621	0.367	0.228
W-LDA+SBoWC2	0.883	0.594	0.525	0.853	0.606	0.517	0.763	0.581	0.443	0.705	0.513	0.362	0.655	0.510	0.334
W-LDA+BoW2	0.790	0.556	0.439	0.792	0.549	0.435	0.743	0.543	0.403	0.695	0.488	0.339	0.637	0.464	0.296
W-LDA+SBoWC3	0.761	0.549	0.418	0.891	0.439	0.391	0.837	0.382	0.320	0.769	0.413	0.318	0.727	0.397	0.289
W-LDA+BoW3	0.735	0.536	0.394	0.874	0.428	0.374	0.830	0.374	0.310	0.763	0.373	0.285	0.722	0.333	0.240
ETM+SBoWC1	0.832	0.462	0.384	0.767	0.457	0.351	0.712	0.405	0.288	0.661	0.416	0.275	0.599	0.423	0.253
ETM+BoW1	0.835	0.433	0.362	0.765	0.421	0.322	0.682	0.367	0.250	0.641	0.378	0.242	0.561	0.377	0.212
ETM+SBoWC2	0.888	0.567	0.503	0.803	0.532	0.427	0.801	0.523	0.419	0.721	0.539	0.389	0.677	0.459	0.311
ETM+BoW2	0.873	0.541	0.472	0.797	0.549	0.437	0.755	0.538	0.406	0.718	0.532	0.382	0.637	0.457	0.291
ETM+SBoWC3	0.869	0.492	0.428	0.814	0.508	0.414	0.778	0.471	0.367	0.738	0.376	0.278	0.705	0.393	0.277
ETM+BoW3	0.865	0.413	0.357	0.809	0.405	0.328	0.784	0.359	0.281	0.705	0.321	0.226	0.613	0.313	0.192

Table 4

The topic model experiment on 20news.

20news	$k = 20$			$k = 40$			$k = 60$			$k = 80$			$k = 100$		
metrics	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
W-LDA+SBoWC1	0.877	0.513	0.450	0.698	0.562	0.392	0.581	0.451	0.262	0.501	0.424	0.213	0.493	0.384	0.189
W-LDA+BoW1	0.857	0.509	0.436	0.687	0.553	0.380	0.570	0.438	0.249	0.462	0.405	0.187	0.421	0.367	0.154
W-LDA+SBoWC2	0.910	0.587	0.534	0.777	0.539	0.419	0.696	0.494	0.344	0.625	0.482	0.301	0.555	0.410	0.228
W-LDA+BoW2	0.877	0.568	0.498	0.745	0.514	0.383	0.691	0.453	0.313	0.623	0.417	0.260	0.533	0.367	0.196
W-LDA+SBoWC3	0.923	0.493	0.455	0.782	0.448	0.350	0.699	0.486	0.340	0.667	0.463	0.309	0.627	0.397	0.249
W-LDA+BoW3	0.833	0.488	0.407	0.725	0.413	0.299	0.678	0.463	0.314	0.665	0.420	0.279	0.622	0.333	0.207
ETM+SBoWC1	0.917	0.552	0.506	0.797	0.499	0.398	0.598	0.509	0.304	0.530	0.507	0.269	0.599	0.423	0.253
ETM+BoW1	0.903	0.518	0.468	0.790	0.496	0.391	0.594	0.500	0.297	0.503	0.495	0.249	0.561	0.377	0.212
ETM+SBoWC2	0.947	0.530	0.502	0.820	0.497	0.407	0.671	0.482	0.323	0.614	0.453	0.278	0.477	0.459	0.219
ETM+BoW2	0.903	0.530	0.479	0.807	0.487	0.393	0.646	0.463	0.303	0.580	0.430	0.267	0.435	0.392	0.179
ETM+SBoWC3	0.922	0.512	0.466	0.814	0.479	0.385	0.670	0.472	0.317	0.598	0.418	0.259	0.481	0.365	0.192
ETM+BoW3	0.887	0.483	0.437	0.796	0.448	0.358	0.659	0.437	0.292	0.567	0.381	0.235	0.463	0.325	0.162



(a) AGnews

(b) 20news

Fig. 3. Comparison of SBoWC and BoW performance on different datasets.

3.4. Proof and experiment of SBoWC applicability

SBoWC Performance in Topic Models: To evaluate the applicability and transferability of SBoWC across different topic models, we designed a set of comprehensive comparative experiments. Specifically, we incorporated SBoWC and the conventional BoW into two widely validated topic modeling methods, W-LDA (Nan et al., 2019) and ETM (Dieng et al., 2019a), and conducted experiments on two benchmark datasets: AGnews and 20news. Furthermore, to examine the influence of the number of base terms, we compared different configurations—SBoWC1/BoW1 (4000 terms), SBoWC2/BoW2 (6000 terms), and SBoWC3/BoW3 (8000 terms). The corresponding experimental results are presented in Tables 3 and 4. Additionally, to provide a more intuitive comparison, we visualized the performance of SBoWC on W-LDA using bar charts, as shown in Fig. 3. We will provide a description of the dataset and metrics in Sections 5.1 and 5.2 together.

By comparing the performance of SBoWC and the conventional BoW under different model frameworks, we observe that applying SBoWC to both W-LDA and ETM generally leads to significant improvements in both TU (Dieng et al., 2019b; Nan et al., 2019) and

Table 5
Classification performance comparison of SBoWC, BoW, and TF-IDF.

Dataset	Classifier	Accuracy			F1-Score		
		SBoWC	BoW	TF-IDF	SBoWC	BoW	TF-IDF
20news	Logistic Regression	0.7890	0.6841	0.7334	0.7850	0.6861	0.7321
	SVM	0.7530	0.6345	0.7260	0.7540	0.6352	0.7265
AGnews	Logistic Regression	0.9390	0.8996	0.9154	0.9370	0.8995	0.9152
	SVM	0.9233	0.8932	0.9147	0.9232	0.8931	0.9146

C_V (Röder et al., 2015) metrics. (A detailed description of the selected evaluation metrics will be provided in Section 5.1.) Even in the few instances where SBoWC underperforms BoW on a single metric, the differences are minimal. Furthermore, for the comprehensive evaluation metric TQ , defined as $TU \times C_V$, SBoWC consistently outperforms or matches the performance of traditional BoW across all settings. These results suggest that the enriched features provided by SBoWC can be effectively leveraged by modern neural topic models.

SBoWC Performance in Classification Tasks: To evaluate the quality of document representations generated by SBoWC across different downstream tasks, we conducted comparative experiments against BoW and TF-IDF using two well-established document classification models: Logistic Regression and SVM. All of them use the first 3000 high-frequency words as the basis terms, and the resulting document representation dimension is 3000. The model was run 20 times respectively on 20news and the AGnews dataset with the same parameters and the average value was taken. As shown in Table 5, SBoWC achieved optimal performance on both accuracy and F1-score metrics across both datasets. Although SBoWC was originally developed to enhance text representation in topic modeling, the experimental results demonstrate that the high-quality document representations generated by SBoWC can also strengthen performance in other downstream tasks, showcasing its broad applicability and transferability.

The Impact of Different Numbers of Base Terms: We further investigated how varying the number of SBoWC base terms (4000, 6000, 8000) affects topic model. The results show that increasing the number of base terms generally leads to an upward trend in the TU metric, suggesting that a larger base term set enhances topic diversity. However, the improvement from 6000 to 8000 terms slows down significantly, and in some cases, the TU metric plateaus or even slightly declines. This indicates that once the number of base terms reaches a certain threshold, the marginal benefit of adding more terms diminishes. Moreover, the inclusion of excessive or low-quality terms may introduce redundancy or noise, thereby negatively impacting the quality of topic generation.

For the C_V indicator, a similar pattern is observed. In many cases, increasing the number of base terms from 4000 to 6000 leads to an improvement in C_V , indicating better semantic coherence among topics. However, when the number of base terms is further increased to 8000, the C_V scores for some models slightly decline. This suggests that while a moderate increase in base terms can enhance topic coherence, an excessive number may introduce redundancy or noise, which in turn blurs topic boundaries and reduces their overall distinguishability.

3.5. Theoretical connection to topic modeling

The SBoWC representation serves not only as a preprocessing step but as a theoretical enabler for topic modeling. Our work unifies semantic clustering (SBoWC) and GSWTM into a coherent mechanism—by clustering semantically related words, SBoWC transforms sparse lexical signals (specifically sparse word co-occurrence signals) into dense semantic structures that naturally align with the heavy-tailed distributions captured by the Gamma prior in GSWTM. In this process, SBoWC absorbs low-frequency word semantics while maintaining high-frequency word importance, which also ensures the revealed long-tail semantics remain compatible with the asymmetric geometry captured by the Sinkhorn optimal transport process in GSWTM. This alignment allows dominant semantic clusters to emerge while preserving long-tail variations through fuzzy memberships. Furthermore, the Wasserstein distance in GSWTM operates over these semantically coherent clusters, ensuring that probability transport reflects both semantic similarity and topic asymmetry. Thus, SBoWC and GSWTM form a unified mechanism in which representation learning and probabilistic modeling are intrinsically coupled, rather than a mere stacking of independent components.

4. GSWTM topic model based on the WAE architecture

In the previous section, we elaborated on the construction process of SBoWC. SBoWC reduces the high-dimensional sparse bag-of-words representation into a lower-dimensional, dense representation through word clustering.

As shown in Fig. 4, the GSWTM employs the Wasserstein Autoencoder (WAE) as its core encoding component to compress the dense representations generated by SBoWC into latent vectors. GSWTM adopts the Gamma distribution as a flexible prior, and samples latent variables using the implicit reparameterization trick to facilitate gradient-based learning. Together, these components form a coherent pipeline—from dimensionality reduction and latent encoding to interpretable and robust topic modeling.

4.1. Problem definition

Given a text corpus $\mathbf{D}$, we preprocess it to obtain the input matrix $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_k]$, where each $\mathbf{s}_d \in \mathbb{R}_{\geq 0}^i$ is the SBoWC representation of the d th document.

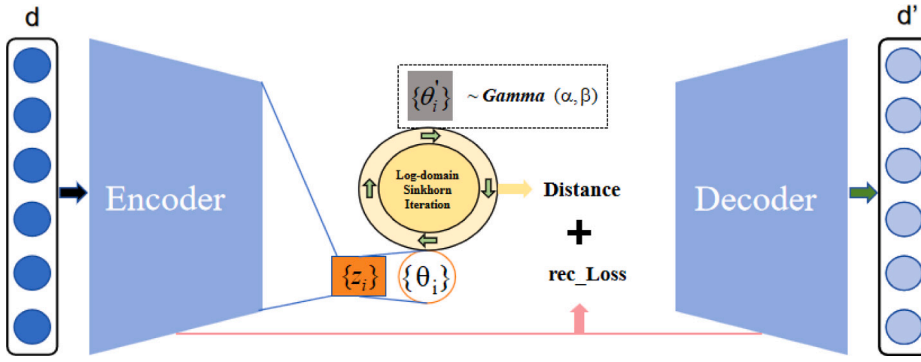


Fig. 4. Structural diagram of GSWTM model.

The goal of GSWTM is to learn low-dimensional latent variables $\mathbf{z}_d \in \mathbb{R}_{\geq 0}^{\kappa}$ that reflect the topic mixture of each document, where κ is the number of topics. This is achieved via an encoder–decoder framework, where the Encoder _{ϕ} maps $\mathbf{s}_d$ to $\mathbf{z}_d$ under a Gamma prior, and the Decoder _{ε} reconstructs $\mathbf{s}_d$, jointly optimized by reconstruction loss and Sinkhorn distance loss. The resulting topic distributions $\{\mathbf{t}_1, \dots, \mathbf{t}_\kappa\}$ yield interpretable top- n words per topic, consistent with the training procedure in Section 4.6.

4.2. GSWTM model architecture

The encoder of the GSWTM model is implemented as a Multilayer Perceptron (MLP), consisting of multiple linear and nonlinear transformation layers. It takes as input the SBoWC representation $\mathbf{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k\}$, where each $\mathbf{s}_d$ denotes the SBoWC vector of the d th document ($d \in \{1, 2, \dots, k\}$). The MLP maps each $\mathbf{s}_d$ to a latent vector $\mathbf{z}_d \in \mathbb{R}^{\kappa}$.

By applying the softmax function to $\mathbf{z}_d$, we obtain the topic distribution vector $\theta_d \in \mathbb{S}^{\kappa-1}$, where each component represents the probability of the document belonging to a particular topic. The encoder is trained to approximate the true posterior $P(\theta | \mathbf{s})$ by learning an inference distribution $Q(\theta | \mathbf{s})$, optimized via the WAE loss. This allows the encoder to generate topic representations that reflect the long-tail characteristics of topic distributions in social media text, as modeled by the Gamma prior.

The decoder maps the topic distribution vector θ_d to a vocabulary-sized output layer with ω units. This layer produces a word probability distribution $\hat{\mathbf{s}}_d$ over the vocabulary, where each value represents the probability of generating a specific word given the topic mixture θ_d .

Encoder: $\mathbf{z}_d = f_{\text{MLP}}(\mathbf{s}_d)$, $\theta_d = \text{softmax}(\mathbf{z}_d)$

Decoder: $p(w_{\omega'} | \theta_d) = \frac{\exp(W_{\text{dec}, \omega'} \theta_d + b_{\text{dec}, \omega'})}{\sum_{\omega''} \exp(W_{\text{dec}, \omega''} \theta_d + b_{\text{dec}, \omega''})}$

Here, $W_{\text{dec}} \in \mathbb{R}^{\omega \times \kappa}$ is the decoder weight matrix, and $b_{\text{dec}} \in \mathbb{R}^{\omega}$ is the bias term. The product $W_{\text{dec}} \theta_d + b_{\text{dec}}$ yields a vocabulary-sized vector of unnormalized word logits, which is then normalized by the softmax function to produce a probability distribution over all vocabulary words.

To better capture the true topic structure of documents in real-world social media text, which often exhibits sparsity, topic intensity imbalance, independence of unexpected events, and long-tail distributions, we adopt the Gamma distribution as the prior for the latent variable $\mathbf{Z}$. Each element $\mathbf{z}_{ij}$ is independently sampled using implicit reparameterization (Knowles, 2015):

$$\mathbf{z}_{ij} = g_{\phi}(u_{ij}; \alpha_{ij}, \beta_{ij}), \quad u_{ij} \sim \mathcal{U}(0, 1)$$

where α_{ij} and β_{ij} are the shape and rate parameters of the Gamma distribution, and g_{ϕ} denotes a differentiable transformation.

4.3. Optimization objective

Following (Tolstikhin et al., 2019), our training objective is to minimize the optimal transport distance between the model-decoded distribution P_{dec} and the empirical data distribution P_w . This leads to the following objective function:

$$\min_{Q_{\theta, \phi}} \mathbb{E}_{w \sim P_w} \left[\mathbb{E}_{s \sim Q_{\theta}(\theta | w)} [C(s, \hat{s})] \right] + \lambda \cdot D(Q_{\theta}(\theta), P_{\theta}(\theta)) \quad (9)$$

where $C(s, \hat{s})$ denotes the reconstruction cost. Similar to common WAE-like models, we also adopt the cross-entropy as the reconstruction loss. It is defined as follows:

$$L_{\text{rec}} = C(s, \hat{s}) = - \sum_{j=1}^{\omega} s_j \log(\hat{s}_j)$$

where $s = (s_1, \dots, s_{\omega})$ denotes the normalized word distribution of the original document over a vocabulary of size ω , and $\hat{s} = (\hat{s}_1, \dots, \hat{s}_{\omega})$ represents the reconstructed word distribution generated by the decoder.

$D(\cdot, \cdot)$ is the divergence between the aggregated posterior and the prior. Common divergence measures such as GAN-based or MMD losses are not suitable for our Gamma prior due to their instability or inability to properly capture the geometry and support of the Gamma distribution. Instead, we adopt the Sinkhorn distance as the divergence term $D(\cdot, \cdot)$, which provides a geometry-aware and numerically stable approximation to optimal transport, better aligning with the support and curvature of the Gamma-distributed latent space.

4.4. Distribution matching

To enhance the flexibility of the optimal transport framework in matching samples from different latent distributions, we refer to the Sinkhorn distance to accommodate non-uniform marginal distributions.

The Sinkhorn distance emerges as the solution to the entropy-regularized optimal transport problem. Given the classical Kantorovich formulation, we introduce entropy regularization to ensure computational tractability:

$$\min_{\pi \in \Pi(\mu, \nu)} \sum_{i,j} \pi_{ij} C_{ij} + \epsilon \sum_{i,j} \pi_{ij} \log \pi_{ij} \quad (10)$$

where $\epsilon > 0$ controls the smoothness of the transport plan. The optimal solution has the form $\pi_{ij}^* = \exp\left(\frac{\alpha_i + \beta_j - C_{ij}}{\epsilon}\right)$, where α_i and β_j are dual variables ensuring marginal constraints.

We employ the log-domain Sinkhorn algorithm to ensure numerical stability when computing the entropy-regularized optimal transport distance. By reformulating the dual variables as $\log u_i = \alpha_i/\epsilon$ and $\log v_j = \beta_j/\epsilon$, we obtain:

$$L_{\text{Sinkhorn}} = \sum_{i,j} \exp\left(\log u_i + \log v_j - \frac{C_{ij}}{\epsilon}\right) C_{ij} \quad (11)$$

where the exponential term represents the optimal transport plan π_{ij}^* . This formulation is derived from the Karush–Kuhn–Tucker optimality conditions: $\frac{\partial L}{\partial \pi_{ij}} = C_{ij} + \epsilon(1 + \log \pi_{ij}) - u_i - v_j = 0$, directly yielding the exponential structure in Eq. (10).

Here, $C_{ij} = d(\theta_i, \theta'_j)$ is the cost matrix defined by a distance function (e.g., cosine or Euclidean), and $\{\theta_i\}_{i=1}^m \sim Q_\theta$, $\{\theta'_j\}_{j=1}^m \sim P_\theta$ denote latent samples drawn from the encoder and prior distributions, respectively. The entropic regularization parameter $\epsilon > 0$ controls the smoothness of the transport plan, with smaller values providing better approximations to true optimal transport but requiring more iterations for convergence.

The dual potentials $\log u \in \mathbb{R}^m$ and $\log v \in \mathbb{R}^m$ are iteratively updated via:

$$\log u_i^{(l+1)} = \log \mu_i - \log \sum_j \exp\left(-\frac{C_{ij}}{\epsilon} + \log v_j^{(l)}\right), \quad (12)$$

$$\log v_j^{(l+1)} = \log \nu_j - \log \sum_i \exp\left(-\frac{C_{ij}}{\epsilon} + \log u_i^{(l+1)}\right) \quad (13)$$

where $\mu = (\mu_1, \dots, \mu_m)$ and $\nu = (\nu_1, \dots, \nu_m)$ are marginal distributions over the source and target samples.

While traditional formulations assume uniform marginals (i.e., $\mu_i = \nu_j = \frac{1}{m}$), we instead allow μ and ν to reflect the probability density values derived from domain-specific prior distributions. This density-based weighting ensures that high-probability regions under the Gamma prior receive appropriate transport resources, and the encoder distribution is properly represented according to the variational inference framework.

In our implementation, the marginal distributions μ and ν used in the Sinkhorn distance computation are defined as follows:

$$\mu_i = \frac{\exp\left(z_{\text{posterior}}^{(i)}\right)}{\sum_{i'} \exp\left(z_{\text{posterior}}^{(i')}\right)}, \quad \nu_j = \frac{p_{\text{Gamma}}\left(\theta_{\text{prior}}^{(j)}\right)}{\sum_k p_{\text{Gamma}}\left(\theta_{\text{prior}}^{(k)}\right)} \quad (14)$$

where μ is the encoder's Softmax-normalized output distribution, and ν is the normalized Gamma prior likelihood at sampled points. This normalization ensures that both μ and ν are valid probability distributions, satisfying the marginal constraints required by the Sinkhorn algorithm.

The choice of Sinkhorn distance over alternatives such as MMD or GAN-based losses is motivated by several factors: (i) *support matching* – Gamma distributions defined on $\mathbb{R}^+$ require geometry-aware distance measures that Sinkhorn naturally accommodates via its cost matrix; (ii) *numerical stability* – unlike adversarial training, Sinkhorn provides more stable and predictable optimization behavior, with convergence properties that are well understood under entropy regularization; and (iii) *geometric preservation* – the optimal transport framework preserves the latent space's underlying structure, which is crucial for learning interpretable representations under Gamma-distributed priors.

4.5. Loss function

To summarize, we define the loss function of our GSWTM model as:

$$\mathcal{L}_{\text{GSWTM}} = - \sum_{\ell=1}^{\omega} s_\ell \log(\hat{s}_\ell) + \lambda \sum_{i,j} \exp\left(\log \mu_i + \log \nu_j - \frac{C_{ij}}{\epsilon}\right) C_{ij} \quad (15)$$

In the actual calculation process, the reconstruction loss is generally several orders of magnitude larger than the regularization term. Therefore, to balance these two terms, we set a scaling factor.

$$\lambda = \frac{1}{\alpha \cdot \log \omega}. \quad (16)$$

Consider a baseline scenario where the document length is d and it contains only a single unique word. Further assume that the output of the decoder is completely uninformative, that is, $\hat{s}_\ell = \frac{1}{\omega}$, $\ell = 1, \dots, \omega$. The end reconstruction loss is $d \cdot \log \omega$. We normalize the reconstruction loss relative to this baseline scenario to 1. Experimental results show that such a choice works well on multiple datasets.

4.6. Topic modeling with GSWTM

The topic modeling based on GSWTM encodes input bag-of-words vectors to generate topic distributions, jointly optimizing reconstruction loss and Sinkhorn distance loss to learn event text topics. The training process of the model is shown in Algorithm 2.

Algorithm 2 GSWTM Model Training

Input: Text corpus $\mathbf{D}$; regularization λ ; number of topics κ ; topic word count n ; regularization coefficient ϵ .

Output: Top- n words $\Psi = [\psi_{t_1}, \dots, \psi_{t_\kappa}]$ per topic, where $\psi_{t_i} \in \mathbb{R}^n$ represents the top- n words for topic i .

Initialize: Encoder parameters ϕ , decoder parameters ξ .

1. Preprocess $\mathbf{D}$ to bag-of-words vectors $\mathbf{S} = [s_1, \dots, s_k]$.

for each epoch do

for each batch $[s_1, \dots, s_a] \subset \mathbf{S}$ **do**

2. Zero gradients.

Forward Pass:

(1). Encode: $\mathbf{Z} = \text{Encoder}_\phi([s_1, \dots, s_a])$.

(2). Sample latent variables $\theta \sim \text{Gamma}(\alpha, \beta)$ via implicit reparameterization.

(3). Sample prior: $\theta'_j \sim \text{Gamma}(\alpha_0, \beta_0)$.

(4). Decode: $[\hat{s}_1, \dots, \hat{s}_a] = \text{Decoder}_\xi([\theta_1, \dots, \theta_a])$.

(5). Compute $\mathcal{L}_{\text{rec}} = -\sum_{i=1}^a s_i \log \hat{s}_i$.

(6). Compute $\mathcal{L}_{\text{Sinkhorn}}$ via Eq (10)-(14) with ϵ .

(7). Compute dynamic λ via Eq (16).

(8). Total loss: $\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{Sinkhorn}}$.

Backward Pass:

(1). Compute $\nabla_\phi \mathcal{L}, \nabla_\xi \mathcal{L}$.

(2). Update ϕ, ξ via Adam.

end for

end for

3. Return top- n words Ψ per topic and model checkpoint.

5. Experiments

5.1. Datasets and preprocessing

To comprehensively evaluate the model's capability under different conditions, we selected four real-world datasets of varying sizes for our experiments (basic attributes of these datasets are listed in Table 6). These datasets span multiple domains of social network textual data, including news platforms, search engines, and online question-answering communities. Below is a detailed description of each dataset:

20news is a widely used benchmark dataset collected and published by Lang (1995). It consists of approximately 20,000 newsgroup documents, evenly distributed across 20 distinct topics, such as computers, technology, entertainment, and sports.

AGnews, constructed by Zhang et al. (2015) is a large-scale news classification dataset. It contains over 120,000 news articles categorized into four classes: World, Sports, Business, and Technology.

Snippet is a short-text dataset proposed by Phan et al. (2008), comprising brief text snippets extracted from web search engine results. These snippets cover a wide range of topics across multiple domains.

Stack Overflow is a real-world dataset provided by Kaggle.¹ and sourced from the Stack Overflow Q&A platform.² It includes textual content such as user questions, answers, and comments, spanning various domains like programming, software development, and related technologies.

¹ <https://www.kaggle.com/stackoverflow/stackoverflow>

² <https://archive.org/download/stackexchange>

Table 6
Dataset statistics after preprocessing.

dataset	Docs	Vocabulary size	Average Length	Class	Basis items
20news	18,828	10,699	83.30	20	6000
AGnews	120,000	11,717	18.65	4	6000
Snippet	12,348	4165	7.06	8	2500
Stack Overflow	19,995	3434	6.73	20	2500

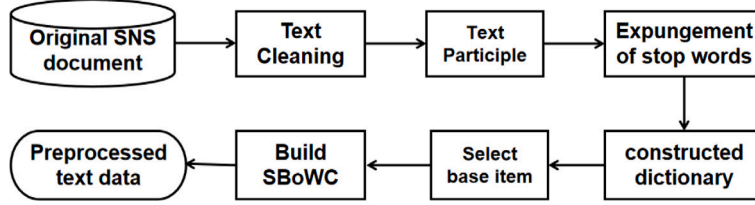


Fig. 5. Data Preprocessing Flow.

Since the original datasets contain noise and irrelevant text, we applied a unified preprocessing pipeline following previous work (Ladani & Desai, 2020; Lee et al., 2021; Li et al., 2022; Liu et al., 2025; Treviso et al., 2023) to ensure reproducibility.

In the data cleaning step, all text was lowercased. A comprehensive regular expression pattern was used to remove numeric characters, punctuation, and special symbols. Tokenization was performed using SpaCy with the “en_core_web_sm” model for English datasets. Stop words were removed based on the standard stop-word list from NLTK.

Vocabulary construction was carried out using the Dictionary class from Gensim, with strict filtering parameters: no_below=20 was used to discard infrequent words, and no_above=0.15 was used to remove overly common terms appearing in more than 15% of the documents. Due to higher lexical diversity and document volume, we retained 6000 basis words for 20news and AGnews, and 2500 for the smaller Snippet and Stack Overflow datasets.

All random processes (e.g., vocabulary sampling and clustering initialization) were controlled with a fixed random seed (random_state=42) to ensure reproducibility. The complete preprocessing workflow is illustrated in Fig. 5.

5.2. Evaluation metrics

While prior work typically selects the top 10–20 words per topic (Hoyle et al., 2021; Lin et al., 2024; Wu et al., 2022), we adopt a compromise approach using the top 15 words as topic descriptors, and we assess topic quality through diversity and coherence using three quantitative metrics.

(1) Topic Diversity

For topic diversity, we evaluate it using Topic Uniqueness (TU) (Dieng et al., 2019b; Nan et al., 2019), which is defined as the proportion of unique vocabulary words among all topic words. The calculation formula is shown in Eq. (18):

$$TU(T) = \frac{1}{T} \sum_{t=1}^T \frac{1}{L} \sum_{l=1}^L \frac{1}{cnt(t, l)} \quad (17)$$

Among T different topics, each topic has L topic words. $cnt(t, l)$ represents the number of times the l th topic word in the t th topic appears in the set of topic words of all topics. The higher the TU value, the higher the topic diversity, where $TU = 1$ indicates all words are distinct.

(2) Topic Coherence

Topic coherence is an indicator for measuring the semantic consistency within a topic. C_V (Röder et al., 2015) is a topic coherence metric based on a sliding window. C_V employs boolean sliding windows and all-pair partitioning of topic words to measure coherence via NPMI-based cosine similarity.

For topic t with n words, C_V computes:

1. NPMI vectors $v_{NPMI}(w_i)$ for each word $w_i \in t$
2. Mean NPMI vector $v_{NPMI}(w_{1:n})$ of all words in t
3. Average cosine similarity between individual and mean vectors

Formally:

$$C_V(t) = \frac{1}{n} \sum_{i=1}^n S_{\cos}(v_{NPMI}(w_i), v_{NPMI}(w_{1:n})) \quad (18)$$

(3) Comprehensive Evaluation

We note that Prior studies (Wu et al., 2020) have observed a trade-off between topic coherence and diversity, where higher TU typically corresponds to lower C_V , and vice versa. To provide a more comprehensive measure for the topic modeling results, we adopt Topic Quality (TQ) as a comprehensive indicator (Dieng et al., 2019b):

$$TQ = C_V * TU \quad (19)$$

TQ comprehensively considers the coherence and diversity of topics. A higher value indicates better topic quality.

5.3. Baselines

To comprehensively evaluate the performance of GSWTM, we conducted extensive experiments by selecting a broad set of strong topic models with diverse methodological advantages. Considering the uncertainties in topic modeling tasks – such as dataset preprocessing, topic number selection, and model adaptability – some traditional methods may outperform recent SOTA approaches in certain cases. Therefore, we included as many competitive baselines as possible to ensure a thorough and reliable evaluation.

PRODLDA (Srivastava & Sutton, 2017b): ProDLDA is a variant of the LDA (Latent Dirichlet Allocation) model. It adopts the Variational Auto-Encoder (VAE) framework, approximates the Dirichlet distribution through a Gaussian distribution, and improves the inference process of the traditional LDA model.

NVDM-GSM (Miao et al., 2018): The inference network learns the latent variable representation of documents through a VAE, it selects a multivariate Gaussian as the prior distribution. The generation network uses a Gaussian - Softmax structure to map latent variables to topic distributions, thereby generating the word distribution of documents.

W-LDA (Nan et al., 2019): A neural topic model proposed under the WAE framework, which directly enforces a Dirichlet prior on the latent document- topic vectors. It utilizes the structure of the latent space and applies an information diffusion kernel in minimizing the Maximum Mean Discrepancy (MMD) to perform distribution matching.

ETM (Dieng et al., 2019b): A document generation model that combines traditional topic models with word embeddings. It uses variational inference to learn a shared low- dimensional representation of words and topics, and each word with a categorical distribution.

BAT (Wang et al., 2020): Applies bidirectional adversarial training to neural topic modeling. The proposed BAT establishes a bidirectional projection between the document – topic distribution and the document – word distribution. It uses a generator to capture semantic patterns from text and an encoder for topic inference.

BERTopic (Grootendorst, 2022): It uses the pre- trained Transformer language model (BERT) to generate document embeddings, then applies UMAP for dimensionality reduction, uses HDBSCAN for clustering, and generates topic representations through a category- based TF-IDF process.

vONTSS (Xu, Jiang, et al., 2023): vONTSS uses a von Mises–Fisher (vMF) based variational auto- encoder and optimal transport. When each topic provides several keywords, vONTSS in the semi- supervised setting generates latent topics and optimizes topic keyword quality and topic classification.

FASTopic (Wu, Nguyen, Zhang, et al., 2024): It is a fast, adaptive, stable, and transferable topic model, different from VAE- based or cluster- based methods. It uses optimal transport among document, topic, and word embeddings from pre- trained Transformers to model the topics and topic distributions of documents.

TraCo (Wu, Pan, et al., 2024): is a novel hierarchical topic modeling framework that addresses the construction of semantically coherent topic hierarchies. It introduces a transport plan dependency mechanism to model sparse and balanced topic dependencies, guiding the hierarchical structure formation.

TopicGPT (Pham et al., 2024): A prompt-based topic modeling framework that leverages large language models to generate interpretable topics with natural language labels and descriptions, rather than traditional word distributions.

5.4. Setting

To fairly compare the performance of the GSWTM model with that of the benchmark models, we made unified settings in the experimental design from the following aspects to ensure that all methods are compared under similar conditions.

Document Representation: Except that our GSWTM model uses the SBoWC representation proposed in this study, all other comparative models use the traditional Bag of Words (BoW) representation as input, converting documents into word frequency vectors. But the bag of words base terms of all models remain the same as our SBoWC. The number of these high- frequency words has been detailed in Section 5.1 and remains consistent across all models.

Use of Word Embeddings: During the training phase of this experiment, to ensure consistency at the word embedding level for each model, we uniformly used the word vectors released by Google,³ which were trained on the Google News corpus (with over 100 billion words), as the basis for all models involving word embeddings. For example, when initializing ETM, it uses 300-dimensional word embeddings by default, which are consistent with the pre-trained Word2Vec word vectors of Google News.

TopicGPT Baseline Setting: To incorporate TopicGPT as a baseline, we made the following adaptations: (1) *Topic Number Control*: We adjusted TopicGPT's refinement threshold to output exactly K topics ($K \in \{20, 40, 60, 80, 100\}$) matching other baselines. (2) *Topic-Word Mapping*: We converted TopicGPT's natural language topics into word distributions by extracting the

³ <https://code.google.com/archive/p/word2vec/>

Table 7

Experimental results on AGnews dataset.

AGnews	$k = 20$			$k = 40$			$k = 60$			$k = 80$			$k = 100$		
Metrics	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
ProdLDA	$\dagger 0.917$	$\dagger 0.429$	$\dagger 0.393$	$\dagger 0.737$	$\dagger 0.409$	$\dagger 0.301$	$\dagger 0.613$	$\dagger 0.419$	$\dagger 0.257$	$\dagger 0.527$	$\dagger 0.463$	$\dagger 0.244$	$\dagger 0.412$	$\dagger 0.376$	$\dagger 0.155$
W-LDA	$\dagger 0.790$	$\dagger 0.556$	$\dagger 0.439$	$\dagger 0.792$	$\dagger 0.549$	$\dagger 0.435$	$\dagger 0.743$	$\dagger 0.543$	$\dagger 0.403$	$\dagger 0.695$	$\dagger 0.488$	$\dagger 0.339$	$\dagger 0.637$	$\dagger 0.464$	$\dagger 0.296$
ETM	$\dagger 0.873$	$\dagger 0.541$	$\dagger 0.472$	$\dagger 0.797$	$\dagger 0.549$	$\dagger 0.437$	$\dagger 0.755$	$\dagger 0.538$	$\dagger 0.406$	$\dagger 0.718$	$\dagger 0.532$	$\dagger 0.382$	$\dagger 0.637$	$\dagger 0.457$	$\dagger 0.291$
BAT	$\dagger 0.757$	$\dagger 0.361$	$\dagger 0.273$	$\dagger 0.745$	$\dagger 0.410$	$\dagger 0.305$	$\dagger 0.362$	$\dagger 0.365$	$\dagger 0.132$	$\dagger 0.352$	$\dagger 0.378$	$\dagger 0.133$	$\dagger 0.219$	$\dagger 0.380$	$\dagger 0.083$
NVDM-GSM	$\dagger 0.517$	$\dagger 0.528$	$\dagger 0.273$	$\dagger 0.442$	$\dagger 0.498$	$\dagger 0.220$	$\dagger 0.362$	$\dagger 0.466$	$\dagger 0.169$	$\dagger 0.362$	$\dagger 0.448$	$\dagger 0.162$	$\dagger 0.347$	$\dagger 0.497$	$\dagger 0.172$
BERTopic	$\dagger 0.847$	$\dagger 0.400$	$\dagger 0.339$	$\dagger 0.803$	$\dagger 0.403$	$\dagger 0.324$	$\dagger 0.752$	$\dagger 0.404$	$\dagger 0.304$	$\dagger 0.692$	$\dagger 0.411$	$\dagger 0.284$	$\dagger 0.638$	$\dagger 0.409$	$\dagger 0.261$
vONTSS	$\dagger 0.850$	$\dagger 0.566$	$\dagger 0.500$	$\dagger 0.762$	$\dagger 0.520$	$\dagger 0.396$	$\dagger 0.650$	$\dagger 0.499$	$\dagger 0.324$	$\dagger 0.568$	$\dagger 0.451$	$\dagger 0.256$	$\dagger 0.420$	$\dagger 0.479$	$\dagger 0.201$
TraCo	$\dagger 0.923$	$\dagger 0.422$	$\dagger 0.390$	$\dagger 0.875$	$\dagger 0.410$	$\dagger 0.359$	$\dagger 0.814$	$\dagger 0.429$	$\dagger 0.349$	$\dagger 0.715$	$\dagger 0.481$	$\dagger 0.344$	$\dagger 0.628$	$\dagger 0.408$	$\dagger 0.256$
FASTopic	0.980	$\dagger 0.386$	$\dagger 0.378$	$\dagger 0.870$	$\dagger 0.399$	$\dagger 0.347$	0.837	$\dagger 0.392$	$\dagger 0.289$	$\dagger 0.729$	$\dagger 0.401$	$\dagger 0.252$	$\dagger 0.579$	$\dagger 0.396$	$\dagger 0.229$
TopicGPT	$\dagger 0.970$	$\dagger 0.385$	$\dagger 0.373$	$\dagger 0.865$	$\dagger 0.398$	$\dagger 0.344$	$\dagger 0.820$	$\dagger 0.390$	$\dagger 0.320$	$\dagger 0.720$	$\dagger 0.400$	$\dagger 0.288$	$\dagger 0.570$	$\dagger 0.395$	$\dagger 0.225$
GSWTM	0.927	0.581	0.539	0.897	0.561	0.503	0.824	0.560	0.460	0.751	0.558	0.419	0.667	0.520	0.347

top 15 words from topic descriptions and supporting quotes. (3) We used GPT-3.5-turbo for topic assignment and GPT-4 for topic generation, following the original implementation. Note that these adaptations constrain TopicGPT from its original design, potentially underestimating its actual capabilities.

Experimental Parameter Settings: The main parameters of the topic model include the number of iterations, batch size, number of topics, number of topic words, etc. We referred to the parameter settings of neural topic models in previous studies. To analyze the performance of topic modeling at different granularities, the number of topics was set to 20, 40, 60, 80, and 100 respectively. Each topic was set with 15 topic words, the batch size was 256, and the number of iterations was 100. Considering the particularity of topic evaluation, we compare the best results within the range of iteration times (when the quality of the topic no longer improves after multiple iterations). Consistent with other neural topic models, we used the ADAM optimizer for parameter optimization, with $\beta_1 = 0.99$ and a learning rate of 0.002.

In terms of the neural network structure, we set the encoder architecture of GSWTM to be the same as that of the encoder layers of models such as W-LDA and ProdLDA, with a total of two hidden layers, each with 100 neurons. The Encoder all used ReLU as the activation function. For the Sinkhorn algorithm, following (Patrini et al., 2019), the regularization coefficient epsilon is 10. We do not want this step to take up too much time, so the maximum number of iterations is only set to 100. The cost functions are all using Euclidean Distance. All baseline hyperparameters were set according to the best-performing configurations in their respective publications.

Hardware and Software Configuration: All experiments were conducted on a desktop running Ubuntu 22.04 with an Intel Core i5-12400F CPU (2.50 GHz), 32 GB RAM, and an NVIDIA GeForce RTX 4060 GPU (8 GB VRAM). The software environment included Python 3.8.6, PyTorch 2.4.1 with CUDA 11.8 support (torch2.4.1+cu118), along with standard libraries such as NumPy and scikit-learn.

Statistical Significance Testing: To rigorously evaluate the performance differences between GSWTM and baseline methods, we employ consistent statistical significance testing throughout our experiments. Given the inherent stochasticity in neural topic model training (e.g., random weight initialization, stochastic gradient descent), we conduct 20 independent training runs for each model to capture this variance. Specifically, for each evaluation metric (TU, C_V , TQ in Section 5.5, and F1 scores in Section 5.9), we perform the following procedure:

1. For each model, we collect performance measurements from 20 independent training runs, which reflect the inherent variance in the stochastic optimization process.
2. We apply unpaired two-tailed t-tests (Welch's t-test) between GSWTM and each baseline method, as samples from different models are independent.
3. We employ Bonferroni correction to adjust for multiple comparisons when testing against multiple baselines, maintaining a family-wise error rate of 0.05.
4. Results are considered statistically significant when the adjusted p -value is less than 0.05.

In Section 5.5, statistically significant improvements are marked with superscript notation $\dagger$ in Tables 7–9. In Section 5.9, significant results are indicated with asterisks (*) in Fig. 9.

5.5. Comparative evaluation

We set different numbers of topics k for testing on different datasets. For the AGnews and 20news datasets, we selected $k = 20, 40, 60, 80, 100$ for the experiments. As for the other two smaller datasets, we set their maximum number of topics to $k = 20, 40, 60$. Tables 7 to 9 show the results of topic quality in terms of topic diversity (TU), topic coherence (C_V), and the comprehensive evaluation index (TQ). Results marked with $\dagger$ indicate that GSWTM achieves statistically significant improvements over the corresponding baseline method ($p < 0.05$).

As observed, except for a few cases, GSWTM consistently achieves state-of-the-art performance across all datasets.

LDA variants (such as ProdLDA) rely on Bayesian inference of word topic distribution, which is suitable for situations where the vocabulary distribution is uniform and the document is long, but performs very unstable on media text data with imbalanced

Table 8

Experimental results on 20news dataset.

20news	$k = 20$			$k = 40$			$k = 60$			$k = 80$			$k = 100$		
Metrics	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
ProdLDA	+0.833	+0.425	+0.354	+0.672	+0.433	+0.291	+0.643	+0.425	+0.273	+0.454	+0.414	+0.188	+0.315	+0.399	+0.126
W-LDA	+0.877	0.568	+0.498	+0.745	+0.514	+0.383	+0.696	+0.453	+0.315	0.633	+0.417	+0.264	+0.533	+0.367	+0.196
ETM	+0.903	+0.530	+0.479	+0.808	+0.479	+0.387	+0.627	+0.433	+0.271	+0.551	+0.441	+0.243	+0.432	0.468	+0.202
BAT	+0.770	+0.438	+0.337	+0.720	+0.481	+0.346	+0.593	+0.488	+0.289	+0.558	+0.451	+0.252	+0.249	+0.309	+0.077
NVDM-GSM	+0.900	+0.476	+0.428	+0.683	+0.422	+0.288	+0.641	+0.411	+0.263	+0.354	+0.476	+0.169	+0.267	0.461	+0.123
BERTopic	+0.763	+0.531	+0.405	+0.677	+0.502	+0.340	+0.688	+0.439	+0.302	+0.620	+0.432	+0.268	+0.569	+0.422	+0.240
vONTSS	+0.840	+0.468	+0.393	+0.732	+0.479	+0.351	+0.661	+0.418	+0.276	+0.558	+0.425	+0.237	+0.505	+0.405	+0.204
TraCo	0.940	+0.457	+0.430	0.813	+0.440	+0.358	0.720	+0.456	+0.328	+0.619	+0.465	+0.288	+0.583	+0.486	+0.283
FASTopic	+0.922	+0.560	+0.516	+0.760	+0.471	+0.358	+0.678	+0.465	+0.315	+0.617	+0.488	+0.301	+0.535	+0.467	+0.250
TopicGPT	+0.915	+0.555	+0.508	+0.755	+0.468	+0.353	+0.670	+0.462	+0.310	+0.610	+0.485	+0.296	+0.530	+0.465	+0.246
GSWTM	0.940	0.568	0.534	0.815	0.556	0.453	0.712	0.492	0.350	0.636	0.494	0.314	0.614	0.553	0.340

Table 9

Experimental results on Snippet and Stack Overflow datasets.

Metrics	Snippet									Stack Overflow								
	$k = 20$			$k = 40$			$k = 60$			$k = 20$			$k = 40$			$k = 60$		
	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
ProdLDA	+0.770	+0.635	+0.489	+0.642	+0.662	+0.425	+0.549	0.676	+0.371	+0.490	+0.351	+0.172	+0.470	+0.397	+0.167	+0.423	0.466	+0.197
W-LDA	+0.973	+0.522	+0.508	+0.868	+0.503	+0.437	+0.717	+0.498	+0.357	+0.937	+0.414	+0.388	+0.743	+0.402	+0.299	+0.612	+0.412	+0.252
ETM	+0.880	+0.624	+0.549	+0.747	+0.623	+0.465	+0.571	+0.633	+0.361	+0.590	+0.427	+0.252	+0.352	+0.407	+0.143	+0.241	+0.379	+0.091
BAT	+0.293	+0.659	+0.193	+0.150	+0.640	+0.096	+0.084	+0.639	+0.054	+0.320	+0.445	+0.142	+0.160	+0.434	+0.069	+0.128	+0.400	+0.051
NVDM-GSM	+0.553	+0.564	+0.312	+0.292	+0.636	+0.186	+0.168	+0.643	+0.108	+0.327	+0.428	+0.140	+0.228	+0.452	+0.103	+0.219	+0.453	+0.099
BERTopic	+0.950	+0.465	+0.442	+0.882	+0.422	+0.372	0.819	+0.424	+0.347	+0.787	+0.365	+0.287	+0.673	+0.390	+0.263	+0.608	+0.405	+0.246
vONTSS	+0.657	+0.436	+0.286	+0.468	+0.355	+0.166	+0.383	+0.349	+0.134	+0.617	+0.497	+0.248	+0.395	+0.355	+0.140	+0.304	+0.335	+0.102
TraCo	+0.883	+0.529	+0.467	+0.548	+0.535	+0.293	+0.473	+0.549	+0.260	+0.793	+0.377	+0.299	+0.458	+0.405	+0.185	+0.362	0.507	+0.184
FASTopic	+0.610	+0.643	+0.392	+0.475	+0.673	+0.320	+0.377	0.685	+0.258	+0.643	+0.377	+0.242	+0.475	+0.399	+0.199	+0.298	+0.400	+0.119
TopicGPT	+0.608	+0.640	+0.389	+0.470	+0.670	+0.315	+0.372	0.680	+0.253	+0.640	+0.375	+0.240	+0.470	+0.396	+0.186	+0.295	+0.398	+0.117
GSWTM	0.977	0.666	0.651	0.888	0.678	0.602	0.777	0.660	0.513	0.970	0.647	0.627	0.852	0.453	0.386	0.743	0.463	0.344

Table 10

Real topic word examples (Computer and Healthcare): Top 10 Keywords per Topic.

Method	Top 10 Keywords
GSWTM	Computer: clock, speed, mhz, performance, throughput, latency, bandwidth, transfer, processor, bus Healthcare: disease, brain, aging, cognition, development, memory, risk, levels, prevention, health
W-LDA	Computer: windows, drive, dos, <u>simplified</u> , scsi, mb, disk, file, ide, pc Healthcare: health, medical, public, care, policy, disease, diagnosis, vaccine, patient, <u>operations</u>
NVDM-GSM	Computer: drive, <u>universal</u> , apple, scsi, mb, video, ide, bus, mhz, controller Healthcare: medical, patient, disease, drug, care, hospital, diagnosis, treatment, vaccine, <u>daily</u>
ETM	Computer: scsi, xterm, address, access, directory, driver, drive, mac, display, <u>basic</u> Healthcare: patients, doctor, treatment, drug, disease, <u>standard</u> , diagnosis, hospital, care, surgery
BAT	Computer: ftp, scsi, drivers, package, hd, ram, processor, bus, <u>basic</u> , ram Healthcare: disease, treatment, <u>daily</u> , patient, drug, doctor, hospital, prevention, vaccine, treatment
vONTSS	Computer: mhz, simms, ram, clock, <u>ordinary</u> , motherboard, chip, memory, cpu, speed Healthcare: doctor, pain, patient, medical, disease, cancer, drug, treatment, diet, <u>normal</u>
FASTopic	Computer: dos, graphics, mac, software, display, windows, video, server, version, system Healthcare: care, patient, doctor, disease, treatment, drug, medical, hospital, diagnosis, <u>daily</u>

distribution. Models such as NVDM-GSM use VAE modeling, which not only has high computational costs but also tends to generate “averaged” results due to KL divergence, resulting in less prominent effects on the AGnews and Stack Overflow.

The professional dataset Stack Overflow contains a large number of technical terms and abbreviations, which makes it difficult for general embedding models to learn accurate semantic representations. For example, ETM and vONTSS perform extremely poorly on Stack Overflow because they rely on pre trained embedded models that cannot handle short texts in technical fields well, resulting in unstable topic distribution. And our model effectively addresses this through the preceding SBoWC text representation.

We note that TopicGPT’s lower performance stems from the fundamental mismatch between its prompt-based generation paradigm and our statistical evaluation framework—the forced topic number specification and word distribution mapping constrain its original design of generating interpretable natural language topics.

It is worth noting that in a few cases, the C_V of GSWTM is slightly lower than that of individual benchmark models. This is not due to the insufficient topic coherence of GSWTM, but rather to the extremely low thematic diversity generated by some baseline models. Due to the mutually exclusive relationship between TU and C_V , when the themes generated by the model are too concentrated and lack diversity, their C_V values may be relatively high. So we introduce TQ indicators to more comprehensively measure the thematic quality of the model.

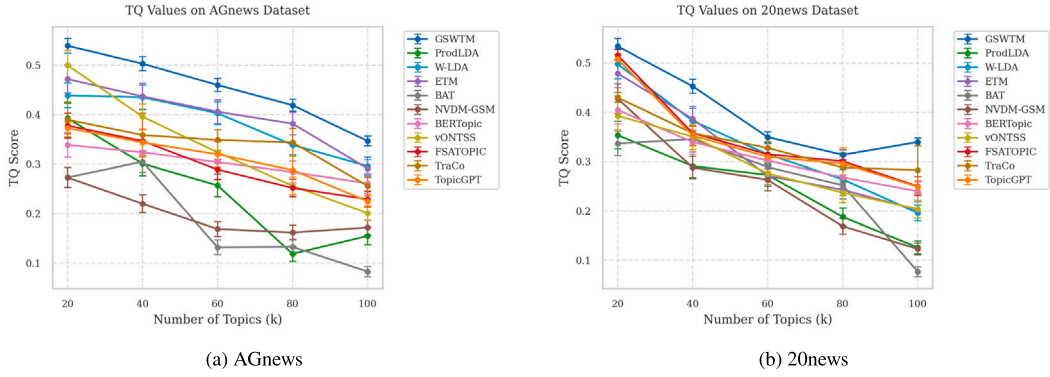


Fig. 6. TQ results with error range.

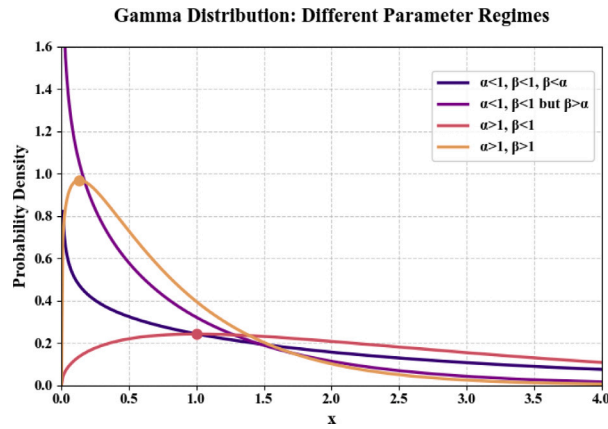


Fig. 7. Gamma distribution diagram.

In our experimental results, GSWTM not only maintains high TU and C_V values on all datasets but also consistently outperforms other baseline models in terms of the TQ indicator. The case of $k = 100$ on the 20news dataset is an exception, where GSWTM unfortunately falls behind TraCo and ranks second. This may be because TraCo leverages hierarchical sharing of statistical information, thereby improving its performance in multi-topic scenarios.

In Fig. 6, we draw the TQ variation trend and error range of the model on the two data sets. It can be seen that GSWTM always maintains a high TQ value under different settings of the number of topics k . The TQ metric further validates the advantages of GSWTM, it can not only generate more diverse topics, but also maintain strong coherence.

Additionally, we provide Table 10 for visual comparison, which displays the top 10 topic words discovered by GSWTM and baseline methods in the two thematic domains of **computer** and **healthcare** using the 20news dataset. Underlined words indicate neutral terms with weak domain specificity. From the table, it can be clearly observed that in the computer domain, terms extracted by GSWTM – such as “clock”, “performance”, and “latency” – are closely semantically related to hardware performance and data transmission. Similarly, in the healthcare domain, words like “disease”, “brain”, and “prevention” focus coherently on physiological functions and disease prevention, with no logical inconsistencies. In terms of topic boundaries, GSWTM’s keywords do not incorporate neutral or interfering terms such as “daily” or “basic”, thus accurately defining the thematic scope. In contrast, baseline methods exhibit noticeable semantic dispersion or boundary ambiguity. This further demonstrates the superior topic quality achieved by GSWTM.

5.6. Gamma parameters analysis

Fig. 7 illustrates the probability distribution of the gamma distribution under different parameter configurations. The selection of gamma distribution parameters significantly influences the performance of topic models (Mazarura et al., 2020). Pre-experimental results indicate that when the shape parameter $\alpha < 1$, the gamma distribution more effectively captures the sparsity characteristics of topic distributions in social network texts. A smaller α value enhances the modeling capability for low-frequency topics, thereby improving the overall expressiveness of the model. Additionally, the scale parameter β must be set to less than 1 and simultaneously

Table 11

Ablation experiment of Gamma parameter.

Gamma	Parameter		AGnews $k = 25$			20news $k = 25$			AGnews $k = 50$			20news $k = 50$		
	α	β	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ	TU	C_V	TQ
gamma1	0.1	0.1	0.812	0.623	0.506	0.784	0.649	0.509	0.812	0.589	0.478	0.765	0.589	0.451
gamma2	0.1	0.3	0.864	0.624	0.539	0.888	0.633	0.562	0.813	0.597	0.488	0.807	0.567	0.457
gamma3	0.1	0.5	0.829	0.661	0.574	0.860	0.636	0.560	0.792	0.581	0.460	0.792	0.571	0.452
gamma4	0.3	0.3	0.927	0.668	0.619	0.901	0.688	0.615	0.809	0.579	0.468	0.763	0.587	0.448
gamma5	0.3	0.5	0.925	0.605	0.560	0.907	0.678	0.615	0.803	0.562	0.451	0.764	0.591	0.452
gamma6	0.3	0.7	0.892	0.679	0.606	0.899	0.675	0.607	0.756	0.571	0.432	0.756	0.559	0.423
gamma7	0.3	0.9	0.912	0.608	0.554	0.885	0.667	0.590	0.813	0.562	0.457	0.763	0.568	0.433
gamma8	0.3	1.0	0.903	0.622	0.562	0.853	0.687	0.613	0.825	0.551	0.455	0.757	0.561	0.425
gamma9	0.5	0.5	0.896	0.622	0.557	0.901	0.646	0.582	0.808	0.551	0.445	0.738	0.561	0.414
gamma10	0.5	0.7	0.907	0.595	0.540	0.932	0.646	0.602	0.807	0.555	0.448	0.747	0.555	0.415
gamma11	0.5	0.9	0.904	0.624	0.564	0.912	0.643	0.586	0.792	0.546	0.432	0.752	0.506	0.381
gamma12	0.5	1.0	0.894	0.631	0.564	0.893	0.650	0.580	0.781	0.502	0.392	0.761	0.504	0.384
gamma13	0.7	0.7	0.840	0.611	0.513	0.864	0.613	0.530	0.791	0.496	0.392	0.732	0.476	0.348
gamma14	0.7	0.8	0.827	0.624	0.516	0.883	0.630	0.556	0.817	0.483	0.395	0.711	0.483	0.343
gamma15	0.7	1.0	0.853	0.593	0.506	0.872	0.637	0.569	0.764	0.489	0.374	0.701	0.509	0.368
gamma16	0.9	0.9	0.875	0.561	0.491	0.891	0.640	0.570	0.773	0.488	0.377	0.703	0.488	0.343

Table 12Performance metrics under different entropy regularization coefficients (ϵ)

ϵ	0.01	0.03	0.05	0.07	0.1	1	10	20	30	50	100
TU	0.9016	0.9066	0.9162	0.9219	0.9167	0.9097	0.9243	0.9187	0.9086	0.9133	0.8973
C_V	0.6237	0.6689	0.6451	0.6397	0.6769	0.6681	0.6611	0.6698	0.6673	0.6784	0.5844
TQ	0.5624	0.6066	0.5911	0.5899	0.6205	0.6077	0.6112	0.6151	0.6062	0.6196	0.5243

smaller than α to ensure that the generated topic distribution maintains appropriate sparsity while preventing reduced topic distinguishability due to over-smoothing.

To determine the optimal parameter combination, we conducted a series of ablation analyses to evaluate the impact of different parameter selections on topic coherence and distribution sparsity. We continued to use TU, C_V , and TQ as evaluation metrics, and carried out comparative analyses under two settings of the number of topics ($k = 25, k = 50$) for the AGnews and 20news datasets respectively. The specific results of the experiment are shown in Table 11.

When α is fixed, a moderate β (such as $\beta = 0.3, 0.5$) usually can obtain a better TQ value, while a too large β (such as $\beta = 1.0$) may lead to a certain improvement in topic coherence (C_V) on some datasets, but it is more at the expense of topic diversity, and the comprehensive evaluation index TQ shows a decline.

When β is fixed, when α is relatively small (such as $\alpha = 0.3$), TQ is usually high; while when α is too large (such as $\alpha = 0.7, 0.9$), a larger α may reduce the diversity and coherence of topics, resulting in a general decline in the evaluation index TQ. For example, for gamma16 ($\alpha = 0.9, \beta = 0.9$), the TQ value on 20news ($k = 50$) is only 0.343, which is much lower than 0.448 of gamma5. In addition, the TQ value of gamma15 ($\alpha = 0.7, \beta = 1.0$) is much lower than other configurations under all dataset settings, which further verifies that a larger α may lead to excessive smoothing of the topic distribution, thereby affecting the model's ability to depict long-tail topics.

The parameters should be configured within the ranges $0.3 < \alpha \leq 0.5$ and $0.3 < \beta \leq 0.9$, subject to the constraint $\alpha \leq \beta$. In our main experiment, we selected $\alpha = 0.3$ and $\beta = 0.3$ as the hyperparameter configuration for the topic model. This parameter combination shows high values of the comprehensive evaluation index TQ under different k values of the two datasets, and can effectively balance the sparsity and coherence of topics.

5.7. Investigation of the regularization coefficient ϵ

To analyze the impact of the entropy regularization coefficient ϵ on the transport plan and computational efficiency, we conducted a systematic evaluation using the 20news dataset with $k=25$ topics. We examined the model's behavior across different regularization strengths $\epsilon \in \{0.01, 0.03, 0.05, 0.07, 0.1, 1, 10, 20, 30, 50, 100\}$. For each configuration, we perform 20 independent runs, each with 100 iterations, to ensure convergence stability. The specific average results are reported in Table 12, and the variation of the selected metrics will be illustrated in Fig. 8. Throughout these experiments, we fixed the Gamma prior parameters at $\alpha = 0.3$ and $\beta = 0.3$ to isolate the effect of ϵ .

From the experimental results, we observe that the evaluation metrics TU and C_V remain remarkably stable across variations of the regularization parameter ϵ . Only when ϵ reaches 100 (losing the geometric advantage of OT) do we notice a marginal performance degradation for both metrics. Our method enables the use of a much smaller ϵ (e.g., < 0.1) to closely approximate the original OT distance. Notably, prior methods (Chen et al., 2025; Liu et al., 2022; Zhao et al., 2021) generally employ ϵ values in the range of 10 to 30, which limits their approximation accuracy and computational performance.

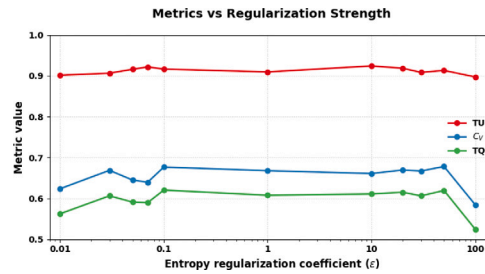
Fig. 8. Metrics vs Regularization coefficients ϵ Strength.

Table 13

Performance comparison of different optimal transport methods on News datasets.

Method	k = 20			k = 100		
	Results	Convergent Epochs	Time per epoch (s)	Results	Convergent Epochs	Time per epoch (s)
Log-Domain Sinkhorn	0.936/0.579	48	20.037	0.574/0.478	59	21.212
Standard Sinkhorn	0.863/0.449	93	24.743	0.546/0.446	95	30.493
Sinkhorn Divergence	0.883/0.482	92	24.569	0.554/0.435	97	33.272
MMD	0.868/0.461	76	31.224	0.342/0.415	99	33.584
KL Divergence	0.653/0.527	86	25.676	0.251/0.500	100+	27.897

5.8. Comparative analysis of optimal transport strategies: Ablation and computational cost

To demonstrate the effectiveness of the Log-Domain Sinkhorn strategy and the computational burden of the model, we designed extensive comparative experiments. Table 13 presents the performance comparison of different approximate Optimal Transport strategies on the 20news dataset (20 runs each with $k = 20/100$). For MMD strategies that do not apply gamma priors, we adopted the general Dirichlet prior, while keeping other parameters consistent.

Experiments show that the average results of our Log-Domain Sinkhorn strategy in multiple dimensions are consistently superior to all baseline methods. Beyond superiority in TU and C_V , our method demonstrates exceptional computational efficiency. Our method achieves the shortest per-epoch time (20.037 s/21.212 s), improving upon standard Sinkhorn and Sinkhorn divergence by approximately 20% or more. When accounting for the optimal number of convergence iterations, our computational advantage becomes even more pronounced, with convergence achieved in fewer than 60 iterations.

It is noteworthy that traditional methods such as KL divergence exhibit poor convergence properties, requiring more than 100 iterations to achieve stable results on larger datasets. Although MMD converges in fewer iterations (76–99), it suffers from the highest per-iteration computational cost (31.224 s and 33.584 s). This renders both approaches unsuitable for large-scale applications.

5.9. Extrinsic evaluation: Text classification performance

To evaluate the effectiveness of topic models from an extrinsic perspective, we further conduct text classification experiments as downstream tasks. Specifically, we use the document-topic distributions learned by topic models as document features and train Logistic Regression and Support Vector Machine (SVM) classifiers to predict the class of each document. We perform experiments on the AGnews and 20news datasets with the number of topics k set to 20, 60, and 100, respectively, and use F1 score as the evaluation metric.

As shown in Fig. 9, our method consistently outperforms all baselines across different classifiers and topic numbers. We have conducted thorough statistical significance tests and marked significant results ($p < 0.05$) with asterisks (*) in the figure. Notably, the performance difference between datasets reveals important insights about our method's adaptability. The structural differences between these datasets – 20news contains 20 fine-grained categories while AGnews has only 4 coarse-grained classes – lead to distinct performance patterns. On AGnews, most competing methods achieve relatively high accuracy (approaching 0.85), though our method still delivers the best performance. However, the evaluation on the more challenging 20news dataset with its 20 categories better demonstrates the strength of our approach. Even with this substantially larger number of categories, our method maintains high accuracy and outperforms the second-best method (FastTopic) by more than 0.05 when $k = 20$. Several existing methods clearly struggle to adapt to this multi-class setting, exhibiting noticeably degraded performance.

These results on datasets with different category granularities highlight both the adaptability and superiority of our proposed topic model, validating their effectiveness for downstream applications.

6. Conclusion

In this paper, we propose GSWTM, a neural topic model built upon the Wasserstein autoencoder framework, which integrates a novel semantic representation method called SBoWC. Our approach tackles core challenges in neural topic modeling for social

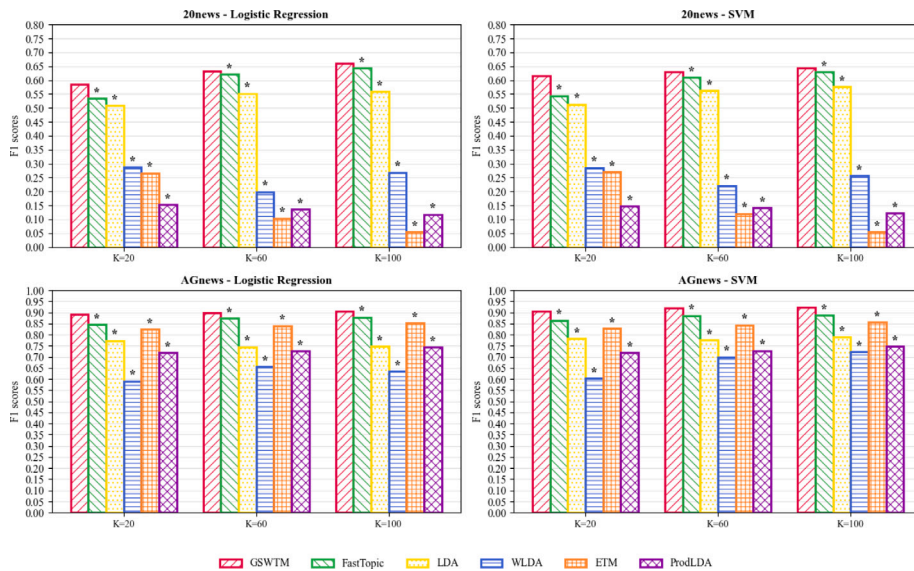


Fig. 9. Performance Comparison of Topic Models on Document Classification.

media texts, including limitations in text representation, long-tail topic distributions, and the approximation of optimal transport. Extensive experiments on four real-world datasets demonstrate that GSWTM consistently achieves strong performance across multiple evaluation metrics.

6.1. Implications and impact analysis

Implications for Modeling: The success of SBoWC confirms that semantic clustering can effectively enhance traditional bag-of-words representations in topic modeling. The use of a Gamma prior offers insights into modeling the sparse and long-tailed distributions common in social media texts. Furthermore, the log-domain Sinkhorn approximation improves the stability of optimal transport computations in neural topic models.

Implications for Architecture Design: The modular nature of our framework offers flexibility in model construction, with SBoWC serving as a transferable representation module across different datasets and tasks. Our end-to-end optimization strategy illustrates that jointly learning representations and topic structures can lead to performance improvements.

Implications for Practical Applications: The model's effectiveness on short and imbalanced texts makes it suitable for real-world applications such as social media analysis and domain-specific content like technical manuals. Its demonstrated ability to generalize to classification tasks further underscores its versatility beyond topic modeling.

6.2. Limitations and future work

While our method shows promising results, there are several areas for future improvement.

SBoWC Representation Considerations: Despite the promising experimental results, the effectiveness of SBoWC is subject to two main limitations. First, it is highly dependent on the quality of the pre-trained word embeddings. General-purpose embeddings may contain inherent biases and often exhibit limited coverage of domain-specific terminology (e.g., technical jargon) or emerging vocabulary, which can directly compromise the accuracy of semantic similarity computations.

Second, although the majority of the computation is performed during the preprocessing phase, constructing the large-scale transformation matrix in SBoWC entails computational and memory requirements that grow quadratically with the size of the vocabulary, representing a potential performance bottleneck. This could be a critical consideration for real-time applications demanding extremely low latency. However, for most offline or non-instantaneous tasks, our experiments demonstrate that the efficiency remains within an acceptable range—even the largest dataset used in our study was processed in under twenty minutes.

GSWTM Model Considerations: The Gamma distribution parameters (α and β) require thoughtful tuning to achieve optimal performance, as demonstrated in our ablation studies. While this provides flexibility in modeling different topic distribution characteristics, it introduces additional hyperparameter complexity. Model convergence is generally stable, but extreme parameter configurations may impact convergence behavior, particularly when α and β values deviate significantly from our recommended ranges.

Regarding computational efficiency, we acknowledge that although our log-domain Sinkhorn approach has significantly improved upon the computational efficiency of standard OT methods, its iterative optimization nature means it requires more computational resources compared to simple divergence metrics—a characteristic that does present potential efficiency challenges

when processing real-world social media datasets with tens of millions of records or more. However, this overhead represents a reasonable trade-off for more accurate distribution matching capabilities: compared to non-OT transport methods, while we sacrifice some temporal efficiency, GSWTM delivers significant improvements in result quality, which has been verified as optimal among methods of the same type. Meanwhile, its overall runtime performance remains within an acceptable range for practical applications and remains competitive with other state-of-the-art approaches.

Future Opportunities: Building on the identified limitations, several directions can be pursued. For SBoWC, future work could explore *multilingual extensions* that adapt semantic clustering across languages, as well as *domain adaptation* through domain-specific pre-trained embeddings to mitigate vocabulary coverage issues. In addition, *scalability enhancements* such as distributed construction of the transformation matrix may help alleviate memory and computation bottlenecks, especially for real-time or large-scale applications.

For GSWTM, promising opportunities include the development of more efficient optimization techniques to further reduce the overhead of iterative OT-based methods, and automated strategies for *hyperparameter tuning* of the Gamma distribution parameters to improve usability without sacrificing flexibility. Moreover, *integration with large language models*, either via embedding fusion or joint training, offers exciting potential to enhance semantic richness and robustness.

CRedit authorship contribution statement

Yi Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Junhao Zhao:** Writing – review & editing, Writing – original draft, Software, Investigation. **Haoran Xu:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Ronghua Zhang:** Visualization, Validation, Investigation, Data curation. **Changzheng Liu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Limengzi Yuan:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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